COMPUTATIONAL METHODS IN
CIVIL ENGINEERING

ENVIRONMENTAL IMPACT DATA ANALYSIS

DELHI–GHAZIABAD–MEERUT RRTS PROJECT

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CIVIL ENGINEERING ASSIGNMENT

TABLE OF CONTENTS

PROJECT OVERVIEW

The Delhi–Ghaziabad–Meerut Regional Rapid Transit System (RRTS) is India's first semi-high-speed rail corridor, covering about 82 km. Implemented by the NCRTC, it is designed to provide fast, reliable, and sustainable regional mobility in the NCR.

Key Objectives:

- Reduce road congestion and travel time.
- Connect Delhi, Ghaziabad, and Meerut in under 1 hour.
- Lower vehicular emissions, improving air quality.
- Promote sustainable urban transport.

RELEVANCE OF DATA

Air Quality Impact

Monitor dust and emissions during construction and operation to safeguard public health and reduce environmental risks.

Noise Pollution

Assess construction and operational noise levels to protect nearby residents and comply with environmental norms.

Traffic Diversion

Track traffic flow changes to evaluate congestion reduction and ensure smooth mobility.

METHODOLOGY

Air Quality Monitoring

Air quality parameters such as PM2.5, PM10, NO_2, and CO were continuously measured using Continuous Ambient Air Quality Monitoring Stations (CAAQMS) installed at selected points along the project corridor. This ensures accurate tracking of pollution levels during both construction and operational phases.

Noise Quality Assessment

Noise levels were recorded with Portable Sound Level Meters at busy junctions and sensitive zones such as residential and institutional areas. Data was collected for both daytime and nighttime to assess variations and compliance with CPCB standards.

Traffic Studies

Traffic surveys were carried out through Classified Traffic Volume Counts (CVCs) at major intersections before and after construction. The data helps evaluate changes in traffic flow, congestion levels, and modal shifts resulting from the RRTS project.

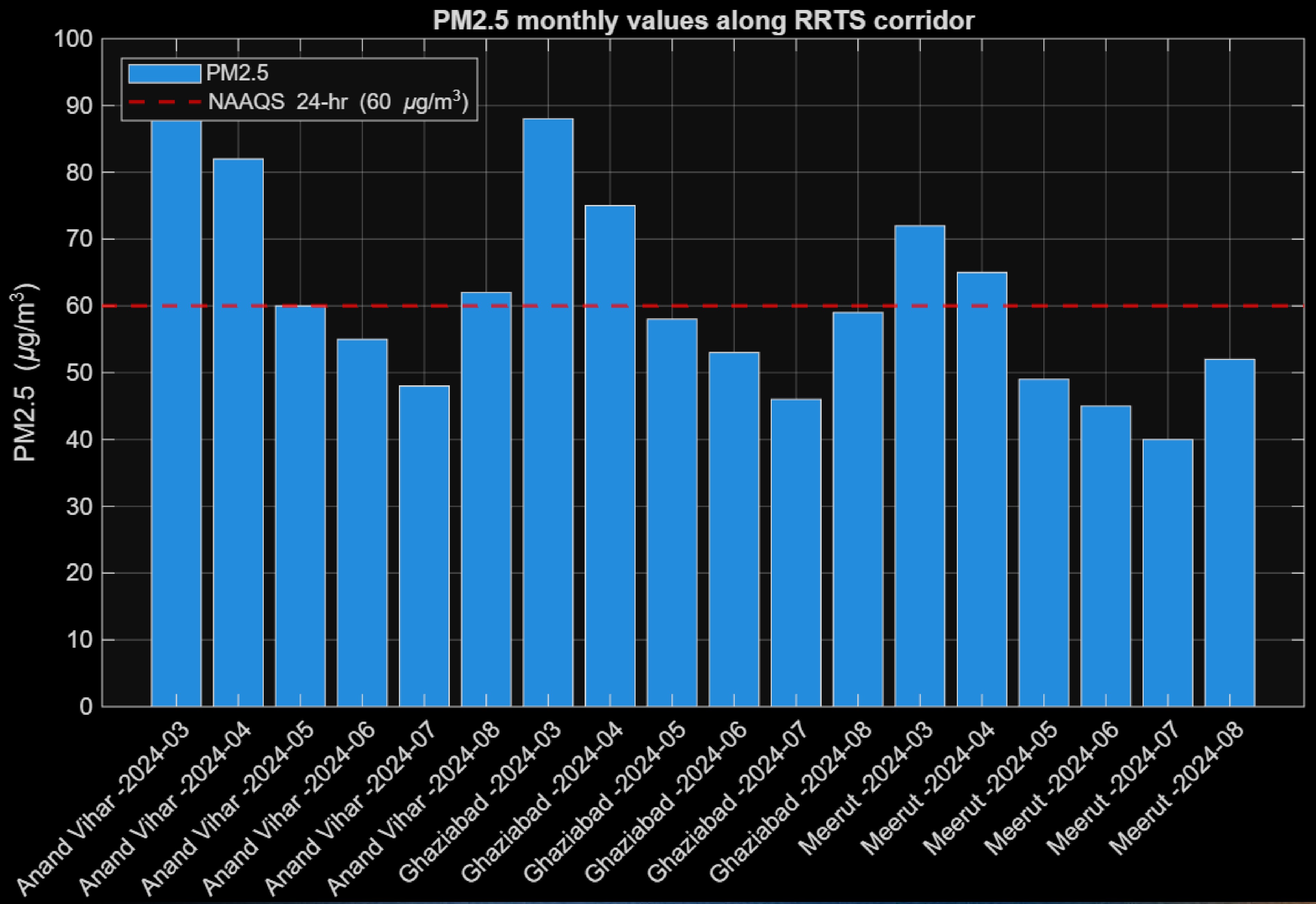
SOURCE OF DATA

- CENTRAL POLLUTION CONTROL BOARD (CPCB)
- MINISTRY OF ENVIRONMENT, FOREST AND CLIMATE CHANGE (MOEF&CC)
- NATIONAL CAPITAL REGION TRANSPORT CORPORATION (NCRTC)
- ASIAN DEVELOPMENT BANK (ADB)
- DELHI POLLUTION CONTROL COMMITTEE (DPCC)
- NATIONAL GREEN TRIBUNAL (NGT)

All sources are verified government and institutional portals ensuring authenticity and reliability of data.

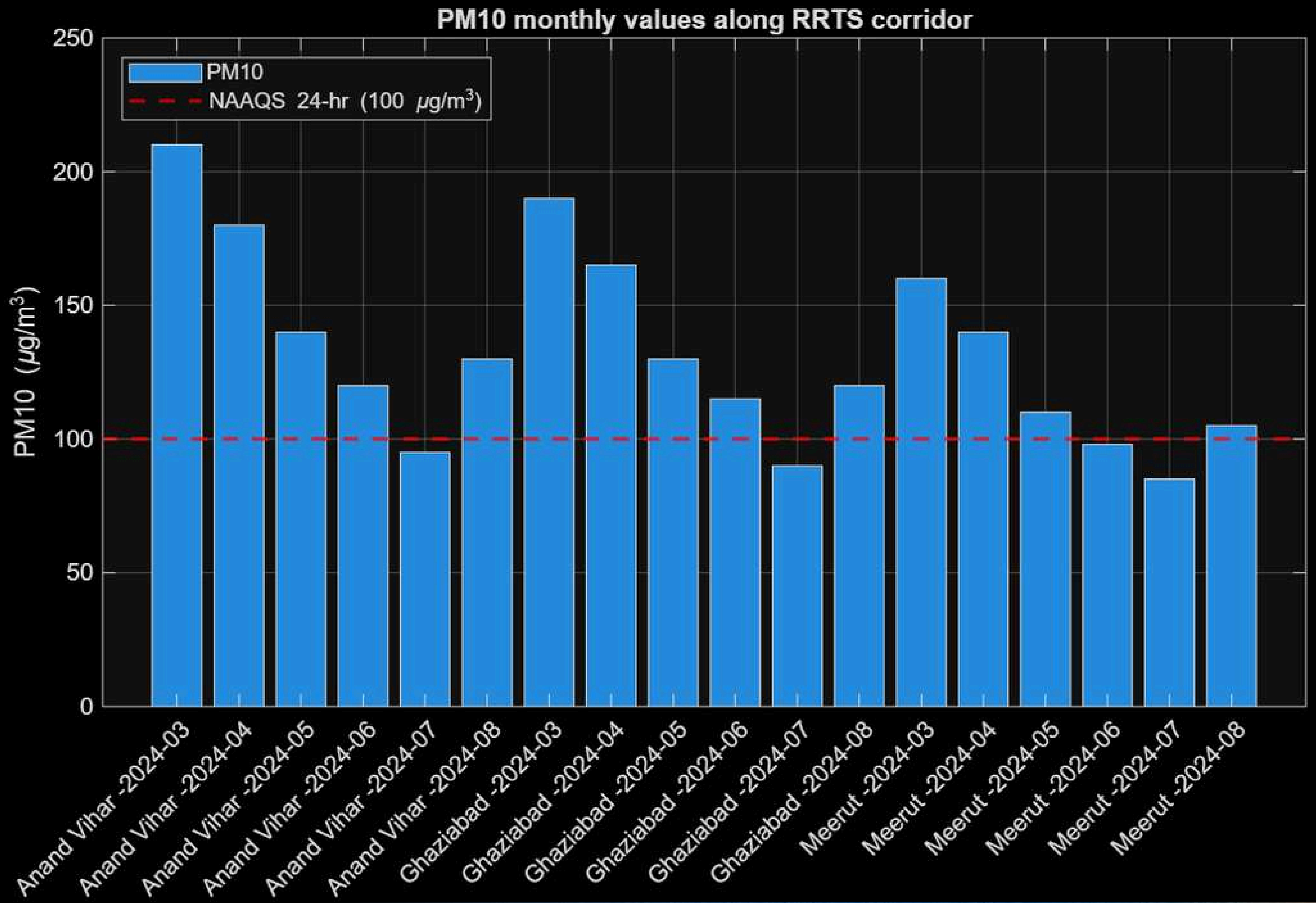
AIR QUALITY PARAMETER (PM 2.5)

```
figure('Name','PM2.5 levels (ug/m3)');
bar(pm25);
xticks(1:length(pm25));
xticklabels(labels);
xtickangle(45);
ylabel('PM2.5 (\mug/m^3)');
title('PM2.5 monthly values along RRTS corridor');
hold on; yline(60,'r--','LineWidth',1.5);
legend('PM2.5','NAAQS 24-hr (60 \mug/m^3)','Location','northwest');
grid on;
```



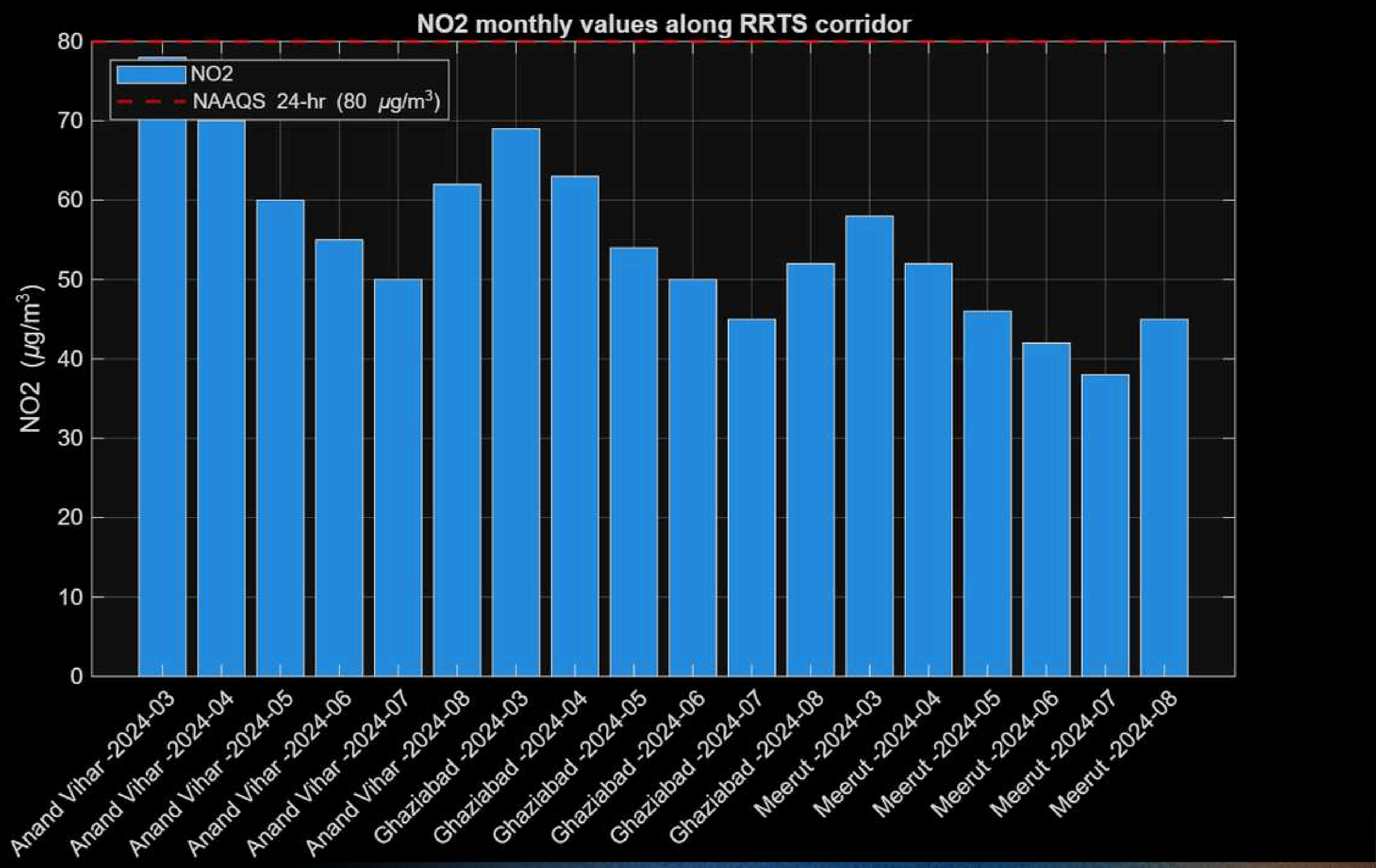
AIR QUALITY PARAMETER (PM 10)

```
figure('Name','PM10 levels (ug/m3)');
bar(pm10);
xticks(1:length(pm10));
xticklabels(labels);
xtickangle(45);
ylabel('PM10 (\mug/m^3)');
title('PM10 monthly values along RRTS corridor');
hold on; yline(100,'r--','LineWidth',1.5);
legend('PM10','NAAQS 24-hr (100 \mug/m^3)','Location','northwest'); grid on;
```



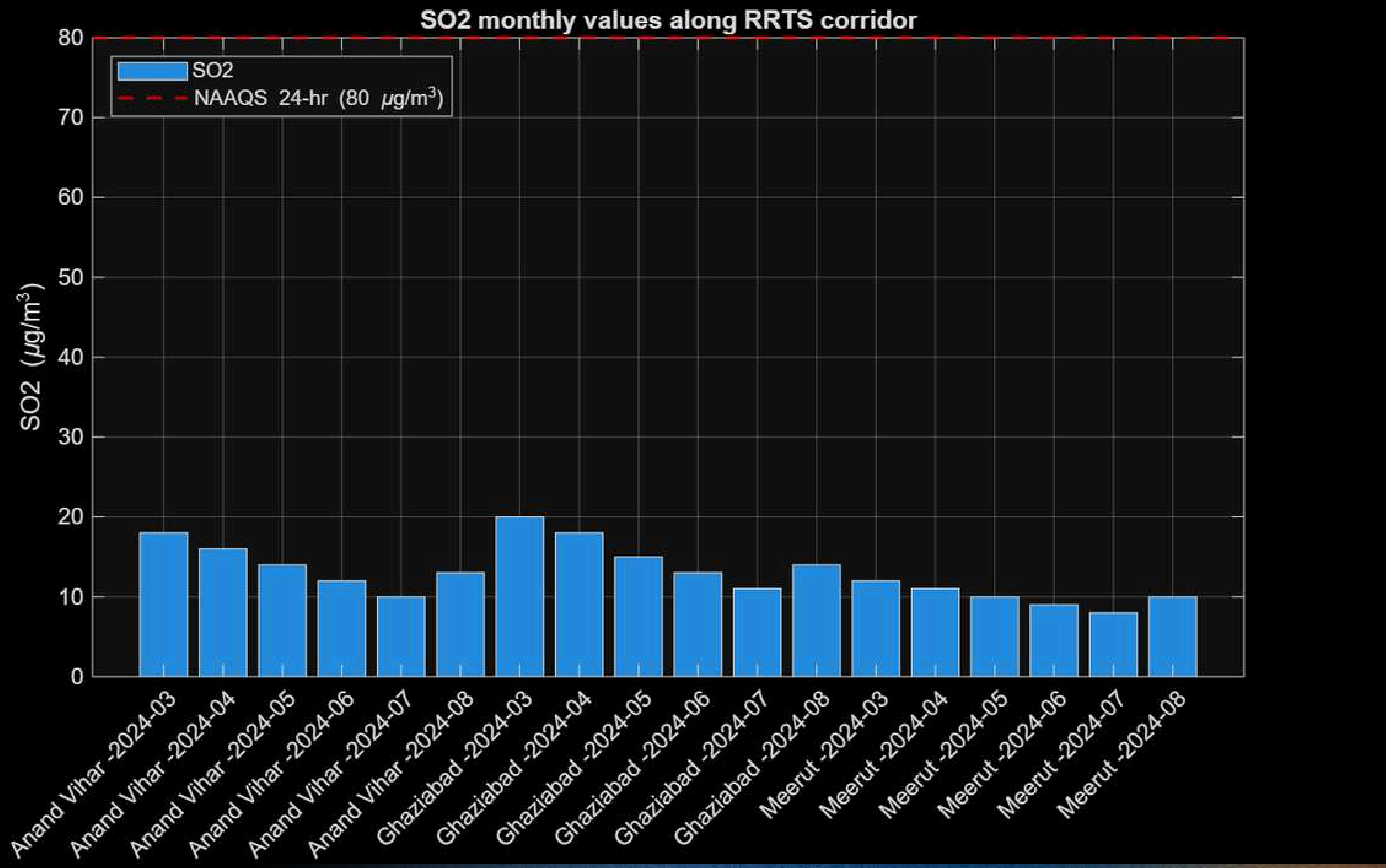
AIR QUALITY PARAMETER (NO2)

```
figure('Name','NO2 levels (ug/m3)');
bar(no2);
xticks(1:length(no2));
xticklabels(labels);
xtickangle(45);
ylabel('NO2 (\mug/m^3)');
title('NO2 monthly values along RRTS corridor');
hold on; yline(80,'r--','LineWidth',1.5);
legend('NO2','NAAQS 24-hr (80 \mug/m^3)','Location','northwest'); grid on;
```



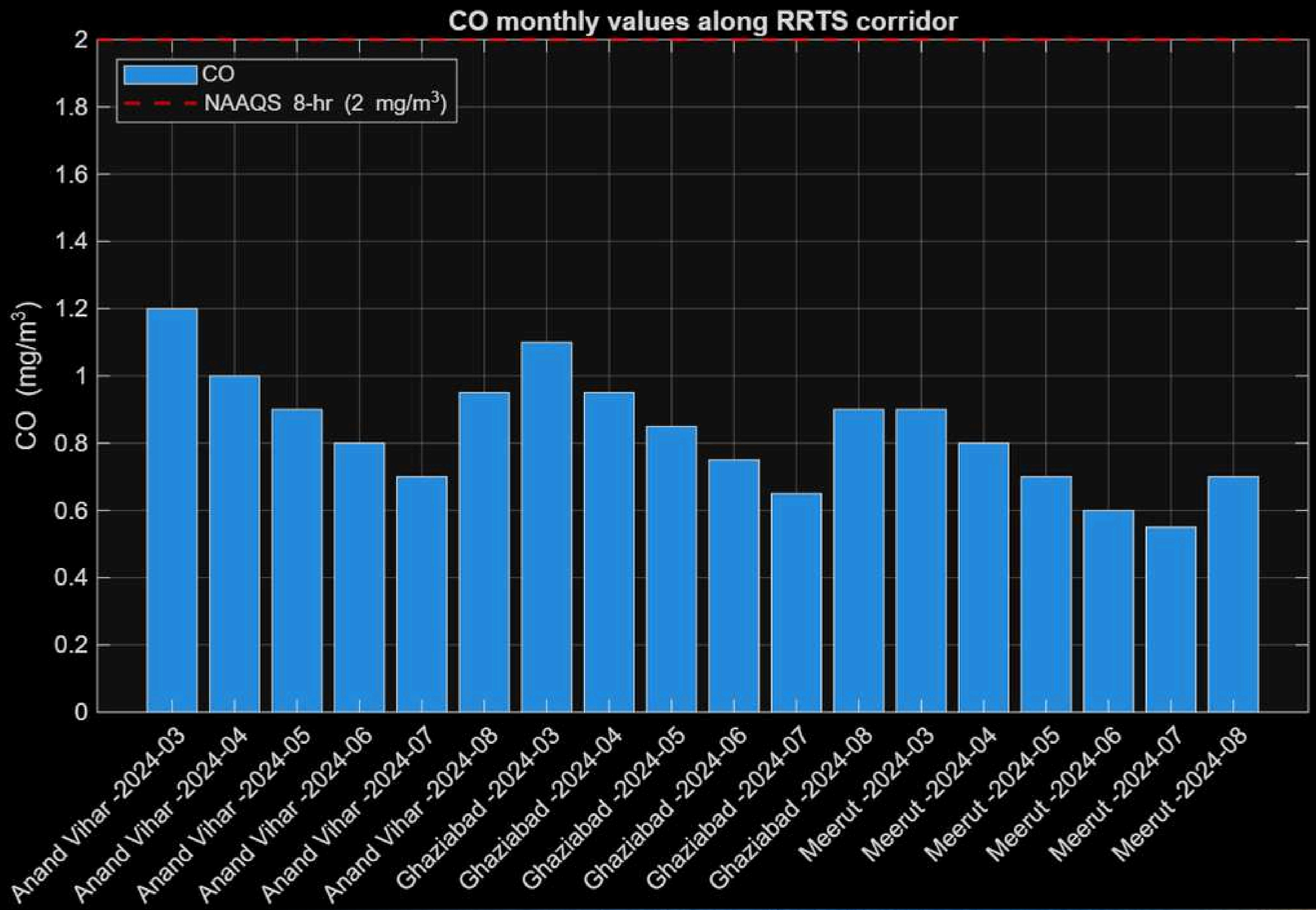
AIR QUALITY PARAMETER (SO2)

```
figure('Name','SO2 levels (ug/m3)');
bar(so2);
xticks(1:length(so2));
xticklabels(labels);
xtickangle(45);
ylabel('SO2 (\mug/m^3)');
title('SO2 monthly values along RRTS corridor');
hold on; yline(80,'r--','LineWidth',1.5);
legend('SO2','NAAQS 24-hr (80 \mug/m^3)','Location','northwest'); grid on;
```



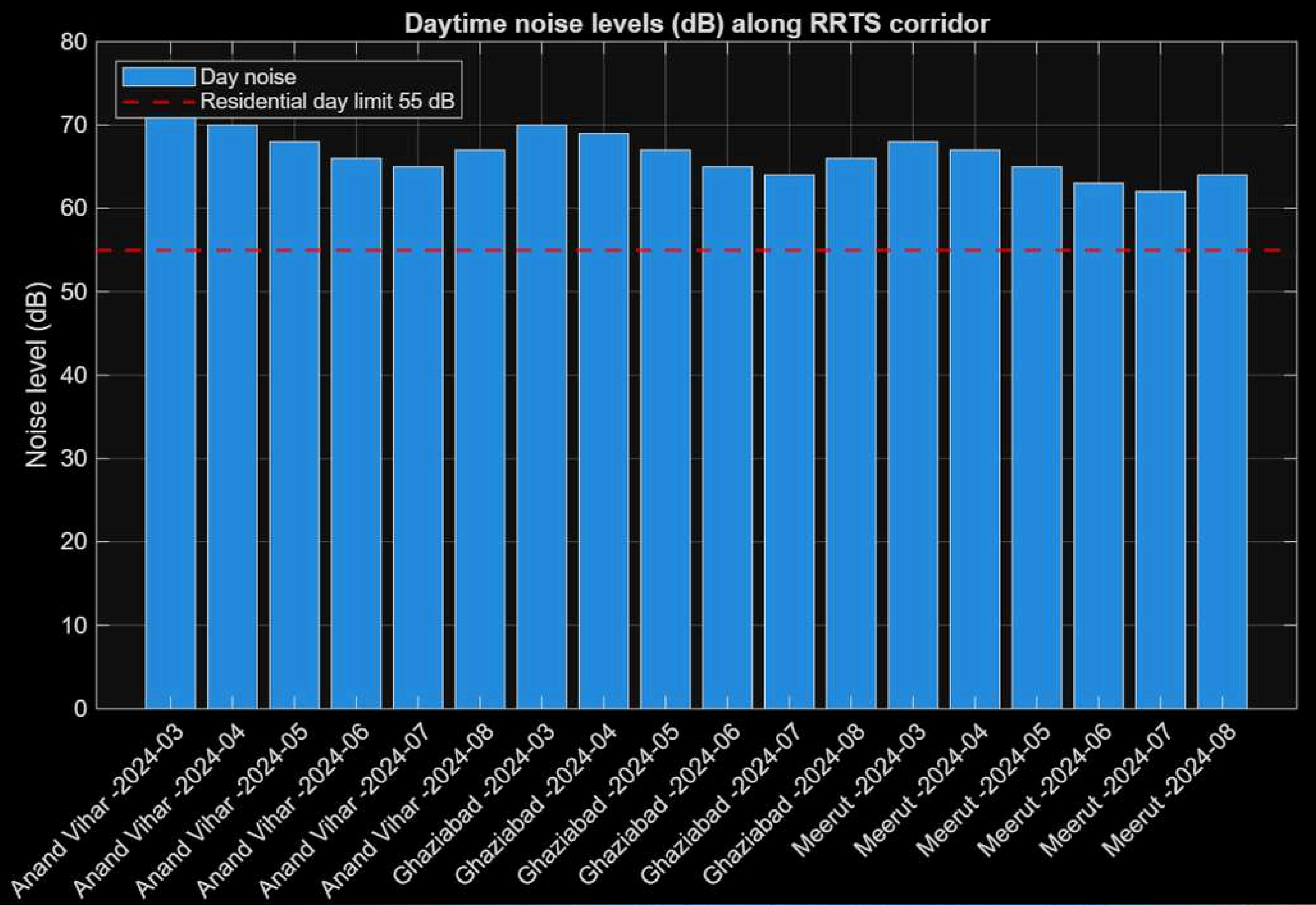
AIR QUALITY PARAMETER (CO)

```
figure('Name','CO levels (mg/m3)');
bar(co);
xticks(1:length(co));
xticklabels(labels);
xtickangle(45);
ylabel('CO (mg/m^3)');
title('CO monthly values along RRTS
corridor');
hold on; yline(2,'r--','LineWidth',1.5);
legend('CO','NAAQS 8-hr (2
mg/m^3)','Location','northwest'); grid
on;
```



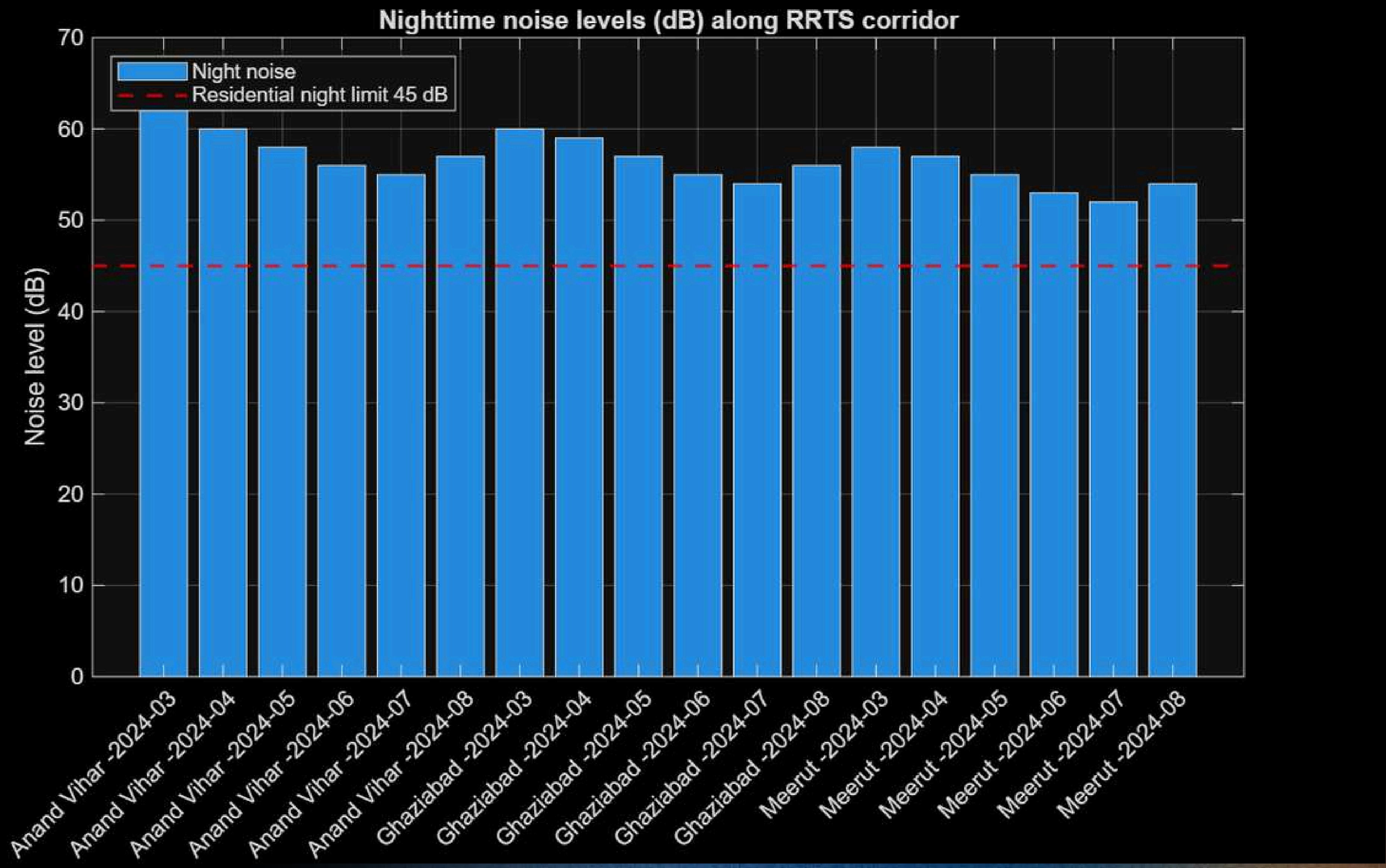
NOISE QUALITY (DAY)

```
figure('Name','Noise Day (dB)');
bar(nday);
xticks(1:length(nday));
xticklabels(labelsN);
xtickangle(45);
ylabel('Noise level (dB)');
title('Daytime noise levels (dB) along
RRTS corridor');
% Residential day limit = 55 dB
hold on; yline(55,'r--','LineWidth',1.5);
legend('Day noise','Residential day limit
55 dB','Location','northwest'); grid on;
```



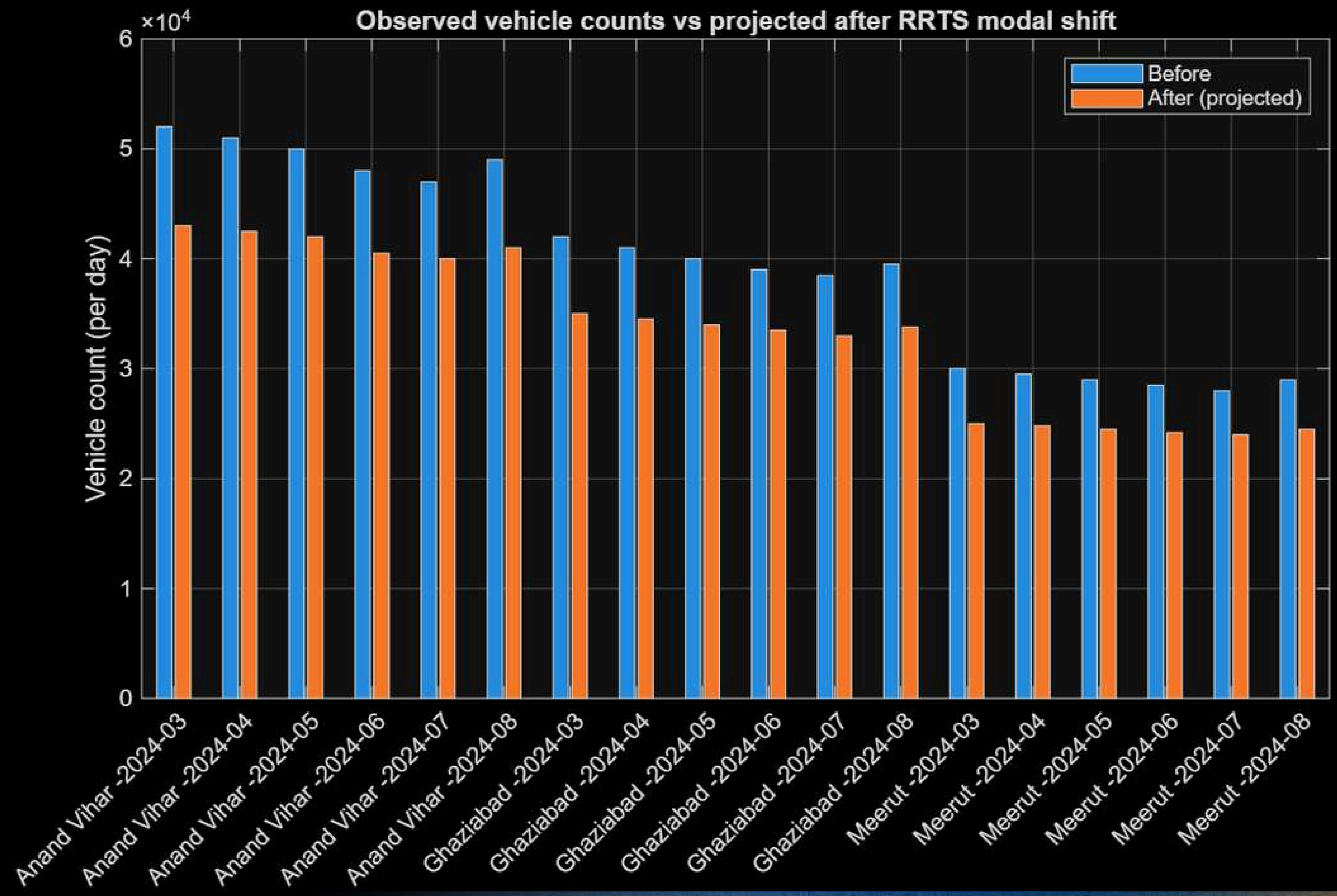
NOISE QUALITY (NIGHT)

```
figure('Name','Noise Night (dB)');
bar(nnight);
xticks(1:length(nnight));
xticklabels(labelsN);
xtickangle(45);
ylabel('Noise level (dB)');
title('Nighttime noise levels (dB) along RRTS corridor');
% Residential night limit = 45 dB
hold on; yline(45,'r--','LineWidth',1.5);
legend('Night noise','Residential night limit 45 dB','Location','northwest'); grid on;
```



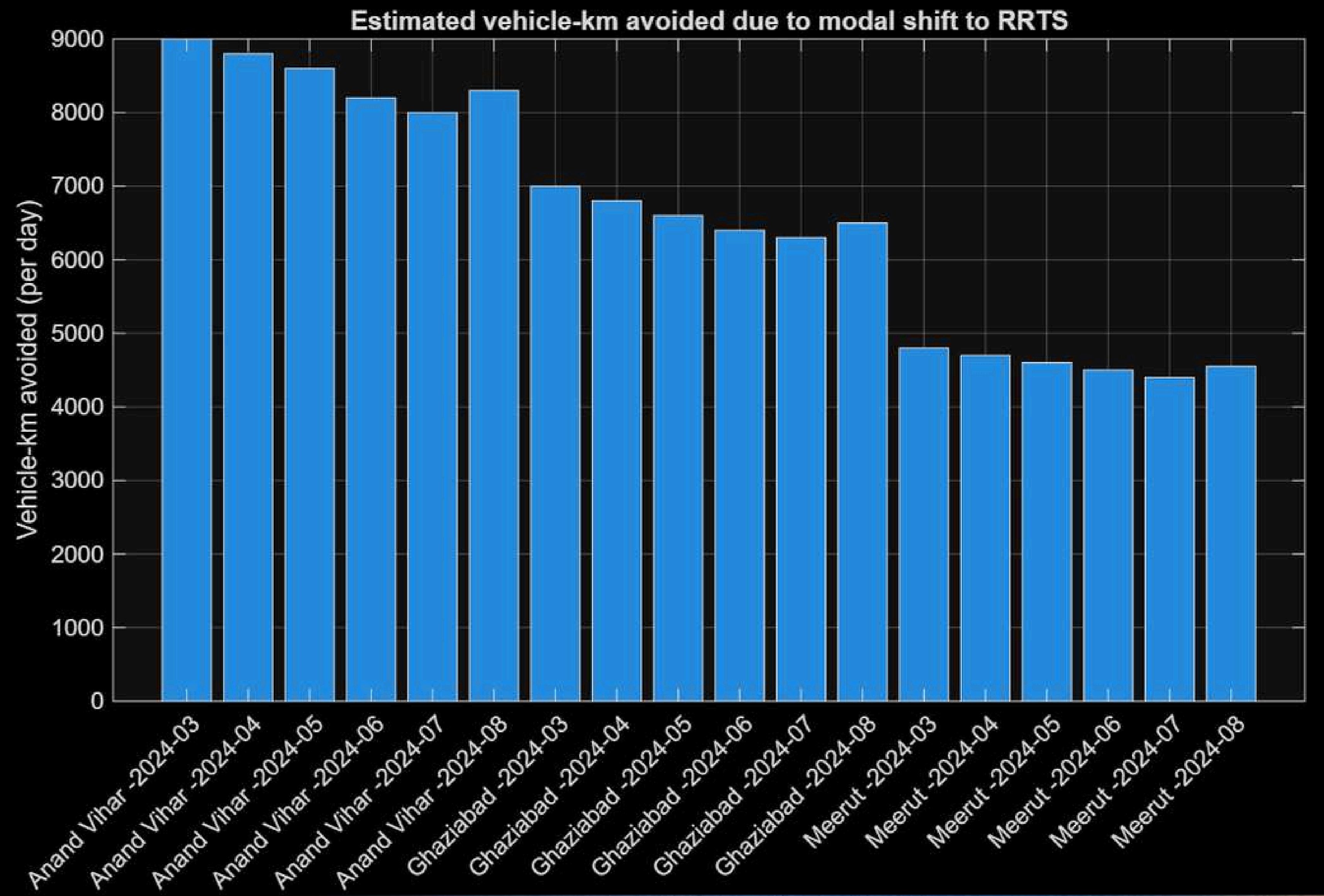
TRAFFIC CHANGE (VEHICLE COUNT)

```
figure('Name','Vehicle counts - before vs after (proj)');
bar([vb va]);
xticks(1:length(vb));
xticklabels(labelsT);
xtickangle(45);
ylabel('Vehicle count (per day)');
title('Observed vehicle counts vs projected after RRTS modal shift');
legend('Before','After (projected)');
grid on;
```



TRAFFIC CHANGE (KM AVOIDED)

```
figure('Name','Vehicle-km avoided per day');
 bar(vkma);
 xticks(1:length(vkma));
 xticklabels(labelsT);
 xtickangle(45);
 ylabel('Vehicle-km avoided (per day)');
 title('Estimated vehicle-km avoided due to modal shift to RRTS');
 grid on;
```



INFERENCES

AIR QUALITY

PM10 and PM2.5 exceeded limits during construction, but operational phase is expected to improve air quality through reduced private vehicle use.

NOISE LEVELS

Higher during construction near sensitive zones, but post-construction levels are projected to comply with CPCB norms.

TRAFFIC FLOW

Significant reduction in private vehicle counts observed, indicating a successful modal shift towards public transport.

Overall Inferences

The monitoring results show that while construction activities temporarily impacted air quality and noise levels, the long-term benefits of the RRTS project are evident in reduced traffic congestion, lower emissions, and improved regional mobility.

DELHI–MEERUT RRTS

A MILESTONE TOWARDS SUSTAINABLE URBAN MOBILITY